

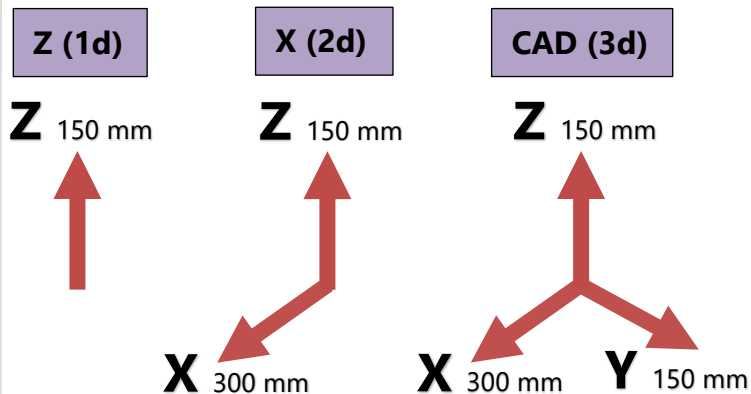


## Technical Data Sheet TCM Eta Nut

Machine with proven single-rotary piston system, powered entirely by electric servo motors.  
Wide range of applications.



Also available with housing



The picture shown shows only one of the almost impossibly diverse variants that can be realized as a customer-oriented Eta.

Call us or contact us so we can bring your idea to a solution.

| Single          | 9, 12, 16, 18, 24                | 9, 12       | Optional                                                            |
|-----------------|----------------------------------|-------------|---------------------------------------------------------------------|
| Boxer           | 9/18, 12/24, 16/32, 18/36, 24/48 | 9/18, 12/24 | For detailed information, see variants further down in the document |
| Liner width     | 45 mm                            | 90 mm       |                                                                     |
| Piston distance | 22.5/45 mm                       | 90 mm       |                                                                     |
| Piston stroke   | 70 mm                            |             |                                                                     |
| Conveyor length | 3 <sup>rd</sup> Party Conveyor   |             |                                                                     |
| Mould widths    | 261 ... 412 mm                   |             | ... 1'200 mm                                                        |
| Weight          | 2'500 kg                         | 3'000 kg    |                                                                     |



## Features

- ✓ Head movement by electric servo motors possible as single-axis (vertical: Z), two-axis (vertical and belt direction of travel: X) or three-axis (CAD)
- ✓ High-precision single rotary piston system with identical dosing volumes (proven for more than 30 years – Single Piston System)
- ✓ No mixing between the pistons possible (1 piston = 1 nozzle)
- ✓ Machine designed for industrial application
- ✓ Quick changeover of different depositing tools (pistons/liner/nozzle plates) without additional tools
- ✓ Number of pistons individually selectable (easy activation / deactivation by using individual couplings)
- ✓ Easy cleaning of the dosing machine without additional tools (Stainless steel or aluminium in ALTEF® coating - for pH values 4 ... 8)
- ✓ Easy cleaning of the pistons and cylinder liners (no tools necessary)
- ✓ Movable touchscreen HMI for machine operation
- ✓ Casting head in free-arm design for easy operation / cleaning / adjustment
- ✓ Fully digital machine control
- ✓ Remote access on PLC, HMI, servo drives possible
- ✓ Software designed for maximum accuracy and flexibility
- ✓ One Shot software already installed on every machine
- ✓ Double-walled hoppers on both sides

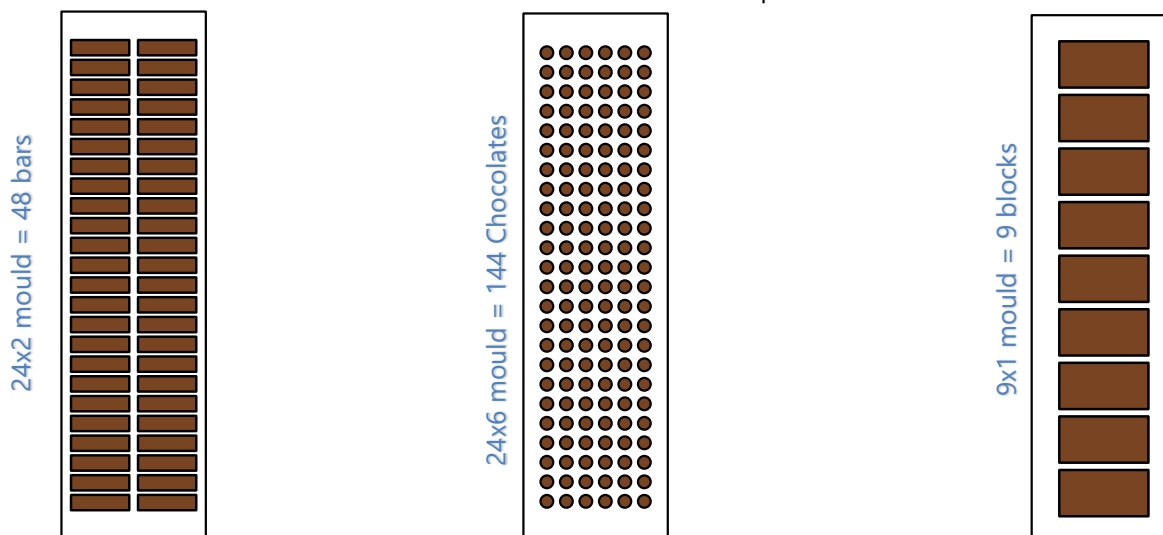
## Options

- For mould widths up to 1200 mm
- Piston lubrication system for one or both piston sides (drip lubrication)
- Different agitator systems for one or both hoppers
- Triple Shot (1 filling / 2 shells OR 2 fillings / 1 shell)
- Quatro Shot (2 fillings / 2 shells)
- Control of downstream machine (e.g. vibration table) possible via control panel
- Y 300 mm (standard 150)
- RFID for mould recognition
- RFID for mould recognition
- Hybrid machine (1 side chocolate, 1 side nut) or as nut machine

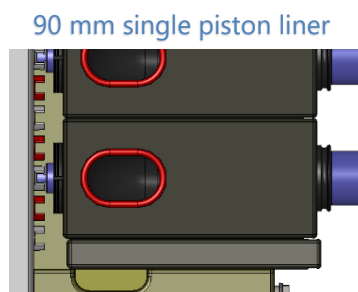
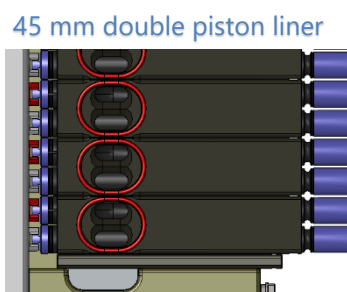
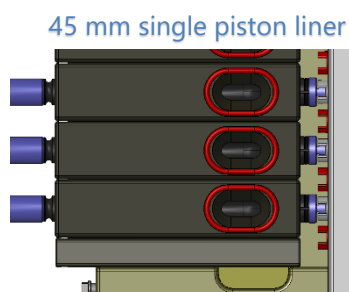


## Combination options with 45 mm cylinder liners

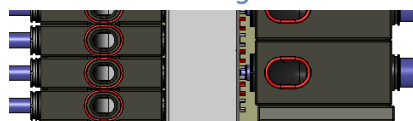
The Eta Nut is designed in such a way that single piston liner in 45 mm, double piston liner in 45 mm and single piston liner in 90 mm can be combined as desired. Here are examples with moulds in the size 830 mm x 230 mm:



The alignment of the off-center pistons, dummies can serve as placeholders.



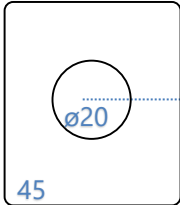
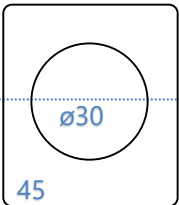
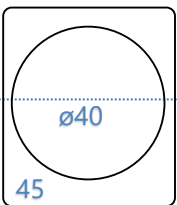
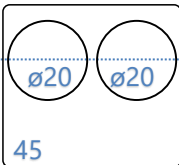
Example of a hybrid solution  
Left 45 mm liner – right 90 mm NI liner



| Piston ø<br>mm | Only<br>Inclusions |    | OneShot or<br>Inclusions |    |    | Piston volume |       |
|----------------|--------------------|----|--------------------------|----|----|---------------|-------|
|                | Nut<br>45          | 90 | Nut-i<br>45/2            | 45 | 90 | ml            | g     |
| 20             |                    |    | X                        | X  |    | 22.0          | 26.4  |
| 30             | X                  |    |                          | X  |    | 49.5          | 59.4  |
| 40             | X                  |    |                          | X  |    | 88.0          | 105.6 |
| 50             |                    | X  |                          |    | X  | 137.4         | 164.9 |
| 60             |                    | X  |                          |    |    | 197.9         | 237.5 |
| 70             |                    | X  |                          |    |    | 269.4         | 323.3 |



## NI (inclusions) piston dimensions (also possible as Hybrid)

|                                          |                                                                                                                                              |                                                                                   |                                                                                                                                                   |                                                                                                                                              |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Liner width 45 mm<br>Piston stroke 70 mm |                                                             |  |                                                                 |                                                           |
| Piston - Ø                               | 20 mm                                                                                                                                        | 30 mm                                                                             | 40 mm                                                                                                                                             | 2x20 mm                                                                                                                                      |
| Volume                                   | 22 ml<br>26.4 g                                                                                                                              | 49.5 ml<br>59.4 g                                                                 | 88 ml<br>105.6 g                                                                                                                                  | 2x22 ml<br>2x26.4 g                                                                                                                          |
| Recommendation                           | One Shot with<br>2 ... 3 mm<br>inclusions =<br>up to 15% share<br><br>Cluster or solid<br>with 4 ... 5 mm<br>inclusions =<br>up to 10% share | Clusters and panels<br>with 4 ... 5 mm<br>inclusions =<br>up to 12% share         | Panels or bars with<br>7 ... 8 mm<br>inclusions =<br>up to 22% share<br><br>Panels or bars with<br>9 ... 11 mm<br>inclusions =<br>up to 25% share | One Shot with<br>2 ... 3 mm<br>inclusions =<br>up to 15% share<br><br>Cluster or solid with<br>4 ... 5 mm<br>inclusions =<br>up to 10% share |

### Example 1: 100g bars with large inclusions

TCM Alpha Nut 24/48-45/2

Piston NI 2x9-45/1-ø40 (2 pistons for one sheet)

18 panels, 15 strokes/min and 60 min/h yields:

➤ **16'200 bars/h or 1'620 kg/h**

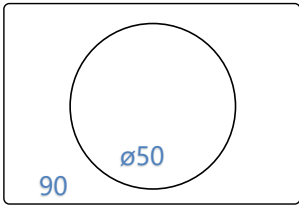
### Example 2: 10g One Shot Chocolate with Small Inclusions

TCM Alpha Nut 24/48-45/2

NI 2x12-45/2-ø20 (1 piston on each side for one chocolate)

24 chocolates, 20 strokes/min and 60 min/h yields:

➤ **28,800 chocolates/h or 288 kg/h**

|                                          |                                                                                                                        |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Liner width 90 mm<br>Piston stroke 70 mm |                                     |
| Piston - Ø                               | 50 mm                                                                                                                  |
| Volume                                   | 137.4 ml<br>164.9 g                                                                                                    |
| Recommendation                           | Panels with 7 ... 8 mm<br>Inclusions = up to 28% share<br><br>Blocks with 9 ... 11 mm<br>inclusions = up to 30%. Share |

### Example 3: 100 g bars with large inclusions

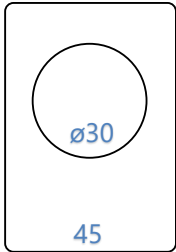
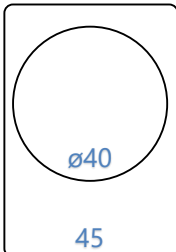
TCM Alpha Nut 24/48

Piston NI 2x9-90-1-ø50 (1 piston for one sheet)

18 panels, 15 strokes/min and 60 min/h yields:

➤ **16'200 bars/h or 1'620 kg/h**

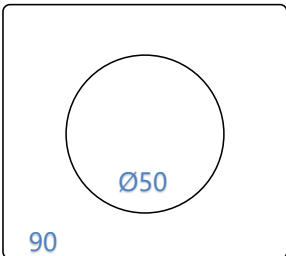
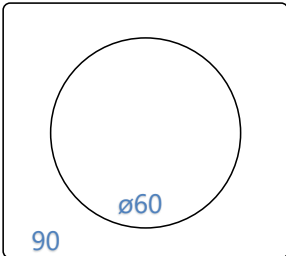
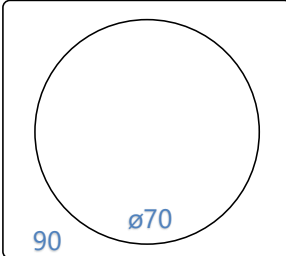
## N (big inclusions) piston dimensions (not possible as Hybrid)

|                                          |                                                                                   |                                                                                                                                 |
|------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Liner width 45 mm<br>Piston stroke 70 mm |  |                                              |
| Piston - Ø                               | 30 mm                                                                             | 40 mm                                                                                                                           |
| Volume                                   | 49.5 ml<br>59.4 g                                                                 | 88 ml<br>105.6 g                                                                                                                |
| Recommendation                           | Clusters and panels with 4 ... 5 mm inclusions = up to 12% share                  | Panels or bars with 7 ... 8 mm inclusions = up to 22% share<br><br>Panels or bars with 9 ... 11 mm inclusions = up to 25% share |

### Example 4: 150g bars with inclusions (mould 830x230)

TCM Alpha Nut 12/24 with piston NI 2x9-45-1-Ø40 (2 pistons for one sheet)  
18 panels, 15 strokes/min and 60 min/h yields:

➤ **16'200 sheets/h or 2'430 kg/h**

|                                          |                                                                                                              |                                                                                                              |                                                                                                              |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Liner width 90 mm<br>Piston stroke 70 mm |                           |                          |                         |
| Piston - Ø                               | 50 mm                                                                                                        | 60 mm                                                                                                        | 70 mm                                                                                                        |
| Volume                                   | 137.4 ml<br>164.9 g                                                                                          | 197.9 ml<br>237.5 g                                                                                          | 269.4 ml<br>323.3 g                                                                                          |
| Recommendation                           | Panels with 7 ... 8 mm inclusions = up to 28% share<br>Panels with 9 ... 11 mm inclusions = up to 30%. Share | Panels with 7 ... 8 mm inclusions = up to 30% share<br>Panels with 9 ... 11 mm inclusions = up to 33%. Share | Panels with 7 ... 8 mm inclusions = up to 33% share<br>Panels with 9 ... 11 mm inclusions = up to 38%. Share |

### Example 5: 200 g blocks of whole nuts (mould 830x230)

TCM Alpha Nut 12/24 with piston NI 2x9-90-1-Ø70 (1 piston for one block)  
18 panels, 15 strokes/min and 60 min/h yields:

➤ **16'200 sheets/h or 3'240 kg/h**

## Applications



- Bars without filling
- Bars with small inclusions up to 3 mm
- Bars with whole nuts



- Filled chocolates in 2 colours (One Shot)
- Filled candy bars in 2 colours (One Shot)
- Filled chocolates in 3 colours (Triple Shot)
- Filled chocolates in 4 colours (Quatro Shot)
- Filled chocolate bars (One Shot Curtain)



- Free forms without filling
- Free forms with fillings
- Free forms with two fillings



- Fillings for Macarons
- Fillings for Alfajores

